2026 Level 2 - Economics

Currency Exchange Rates: Understanding Equilibrium Value

a. calculate and interpret the bid-offer spread on a spot or forward currency quotation and describe the factors that affect the bid-offer spread

b. identify a triangular arbitrage opportunity and calculate its profit, given the bid-offer quotations for three currencies

c. explain spot and forward rates and calculate the forward premium/discount for a given currency

d. calculate the mark-to-market value of a forward contract

e. explain international parity conditions (covered and uncovered interest rate parity, forward rate parity, purchasing power parity, and the international Fisher effect)

f. describe relations among the international parity conditions

g. evaluate the use of the current spot rate, the forward rate, purchasing power parity, and uncovered interest parity to forecast future spot exchange rates

h. explain approaches to assessing the long-run fair value of an exchange rate

i. describe the carry trade and its relation to uncovered interest rate parity and calculate the profit from a carry trade

j. explain how flows in the balance of payment accounts affect currency exchange rates

k. explain the potential effects of monetary and fiscal policy on exchange rates

l. describe objectives of central bank or government intervention and capital controls and describe the effectiveness of intervention and capital controls

m. describe warning signs of a currency crisis

Bid-Ask Spread

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- 2 quoting conventions

B.P = base.price

or $P/_B = \frac{\text{price}}{\text{base}}$

e.g./

USD.CAD = 1.3410

(1)

$\frac{CAD}{USD} = 1.3410$

base = bottom

(1 USD is worth $1.3410)

long ⇒ buy the base by paying the price

- buy 1 USD by paying $1.3410 CAD - now own $1 USD
- long USD/short CAD

short ⇒ sell the base and receive the price

- sell 1 USD by receiving $1.3410 CAD - now own $1.3410 CAD
- short USD/long CAD

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- settlement on spot transactions ⇒ T + 2

EUR.USD or $USD/_{EUR}$

Interbank Market

← spread →

1.0518 bid Dealer offer 1.0519

← spread →

1.0517 bid Client offer 1.0520

then sells for 1.0518

then buys for 1.0519

dealer willing to buy EUR for 1.0517 USD

dealer willing to sell EUR for 1.0520 USD.

client sells EUR for 1.0517 USD (hit the bid)

client pays 1.0520 USD for EUR (pays the offer)

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⇒ Spread depends on

1) Currency pair involved/

EUR.USD or/ USD/EUR
USD.JPY JPY/USD
GBP.USD USD/GBP

very deep & liquid pairs
- very tight spreads

crosspairs

CHF.MXN or/ MXN/CHF
MXN.JPY JPY/MXN

wider spreads

2) Time of Day/
3 big markets
- Japan
- Europe
- N. America

Tokyo	09:00	13:00	17:00	21:00	01:00 Day+1	05:00 Day+1	09:00 Day+1
London	00:00	04:00	08:00	12:00	16:00	20:00	24:00
New York	19:00 Day-1	23:00 Day-1	03:00	07:00	11:00	15:00	19:00

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3. Volatility - esp. right in front of major data releases

8:30am NY
8:29 - 50 sec. 1 pip
8:30.45 20 pips

+4) Who the broker dealer is

- some will pass on interbank bids/asks but charge 1 pip commission

- some will spread the interbank bid/ask but charge no commission

Triangular Arbitrage

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⇒ Dealer bid-offer quotes must respect 2 arbitrage constraints

1) Dealer bid < Interbank offer
Dealer offer > Interbank bid

e.g./
Int.B. 1.0522/1.0524 - Buy @ 24
Sell @ 25
Dealer 1.0525/1.0527

2) Triangular arbitrage
Dealer crossrate bid < implied Interbank crossrate offer

so, if we have $\frac{USD}{EUR}$ & $\frac{JPY}{USD}$, we can imply $^{JPY}/_{EUR}$
(EUR.USD) (USD.JPY) (EUR.JPY)

1.0522/1.0524 118.10/118.15

multiply = crossrate offer

multiply = crossrate bid

Bid = 1.0522 × 118.10 = 124.26 Offer = 1.0524 × 118.15 = 124.34

EUR.JPY ⇒ 124.26/124.34

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⇒ general approach

$$\frac{C}{B} = \frac{A}{B} \times \frac{C}{A}$$

bid = bid$(^A/_B)$ × bid$(^C/_A)$
offer = offer$(^A/_B)$ × offer$(^C/_A)$

$$B.C = (B.A) \times (A.C)$$

bid = bid(B.A) × bid(A.C)
offer = offer(B.A) × offer(A.C)

what about: find $^C/_B$? $\frac{A}{B}$ & $\frac{A}{C}$ ⟶ invert $\left\{ \begin{array}{l} \text{offer} = {}^1/_{\text{bid}} \\ \text{bid} = {}^1/_{\text{offer}} \end{array} \right.$

e.g./ GBP.USD

$\frac{USD}{GBP}$ 1.5644/1.5646

$\frac{USD}{EUR}$ 1.3649/1.3651

EUR.USD

find $\frac{GBP}{EUR}$ (EUR.GBP)

① invert $\frac{USD}{GBP}$ bid = $^1/_{1.5646}$ = 0.63914
offer = $^1/_{1.5644}$ = 0.63922

$\frac{GBP}{EUR}$ bid = .63914 × 1.3649 = 0.8724
offer = .63922 × 1.3651 = 0.8726

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Examples/

USD/EUR	1.4559/1.4561
JPY/USD	81.87/81.89
CAD/USD	0.9544/0.9546
SEK/USD	6.8739/6.8741

1. bid-offer for SEK/EUR
   bid = 6.8739 × 1.4559 = 10.0077
   offer = 6.8741 × 1.4561 = 10.0094

2. bid-offer for JPY/CAD
   Step 1. invert CAD/USD
   bid = $^{1}/_{0.9546}$ = 1.04756
   offer = $^{1}/_{0.9544}$ = 1.04778
   bid = 1.04756 × 81.87 = 85.76
   offer = 1.04778 × 81.89 = 85.80

3. if JPY/CAD were quoted at 85.73/85.75 by a dealer, triangular arb. would involve … Buying CAD from dealer at 85.75
   Selling CAD Intermarket for 85.76

4. What about 85.74/85.81?

Spot vs. Forward Rates

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⇒ forward exchange rates must satisfy an arbitrage relationship

$^{P}/_{B}$ now becomes $^{f}/_{d}$ = $^{\text{foreign}}/_{\text{domestic}}$

1 unit of d

invest at the domestic risk-free rate = i_d
↓
$(1 + i_d)$
end of year payoff

convert to f at $S_{f/d}$ spot
↓
invest at the foreign risk-free rate = i_f
↓
$S_{f/d}(1 + i_f)$
↓
convert to d at $F_{f/d}$ forward

must be the same return

$$\frac{S_{f/d}(1+i_f)}{F_{f/d}} = S_{f/d}(1+i_f)\left(\frac{1}{F_{f/d}}\right)$$

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$$(1 + i_d) = S_{f/d}(1 + i_f)\left(\frac{1}{F_{f/d}}\right)$$

- risk-free rates quoted using Libor
  - lets introduce the day-count convention

$$\left(1 + i_d\left[{}^{Act.}/_{360}\right]\right) = S_{f/d}\left(1 + i_f\left[{}^{Act.}/_{360}\right]\right)\left(\frac{1}{F_{f/d}}\right)$$

divide by multiply by

$$F_{f/d} = S_{f/d}\left(\frac{1 + i_f\left[{}^{Act.}/_{360}\right]}{1 + i_d\left[{}^{Act.}/_{360}\right]}\right)$$

equation called 'covered interest rate parity'

$$\underbrace{F_{f/d} - S_{f/d}}_{\text{forward premium or discount}} = S_{f/d}\left(\frac{{}^{Act.}/_{360}}{1 + i_d\left[{}^{Act.}/_{360}\right]}\right)(i_f - i_d)$$

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⇒ domestic currency will trade at a forward:

premium $\left(F_{f/d} > S_{f/d}\right)$ if $i_f > i_d$

discount $\left(F_{f/d} < S_{f/d}\right)$ if $i_f < i_d$

- now lets put ${}^{f}/_{d}$ back into ${}^{P}/_{B}$: $F_{P/B} = S_{P/B}\left(\frac{1 + i_P\left[{}^{Act.}/_{360}\right]}{1 + i_B\left[{}^{Act.}/_{360}\right]}\right)$

e.g./ $S_{CAD/AUD} = 1.0145$

270 day Libor AUD = 4.87%

270 day Libor CAD = 1.41%

forward premium/discount?

$$F_{P/B} - S_{P/B} = S_{P/B}\left(\frac{{}^{Act.}/_{360}}{1 + i_B\left[{}^{Act.}/_{360}\right]}\right)(i_P - i_B)$$

$$= 1.0145\left(\frac{{}^{270}/_{360}}{1 + .0487\left({}^{270}/_{360}\right)}\right)(.0141 - .0487)$$

$$= -.0254$$

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Maturity	Spot Rate or Forward Points
Spot (USD/EUR)	1.3549/1.3651 possible mistake
One month	-5.6/-5.1 ⇒ 1.3549 1.3651
Three months	-15.9/-15.3 5.6 5.1
Six months	-37.0/-36.3 1.35434 1.36459
Twelve months	-94.3/-91.8

forward fx. rates are quoted in terms of points

Standard forward dates are typically quoted ⇒ clients can select any time period (i.e. non-standard a.k.a. broken)

the longer the term, the wider the bid-offer spread
a) liquidity declines
b) counterparty credit risk increases
c) interest rate risk increases

- rates will be interpolated based on standard settlement dates

Mark-to-Market Value

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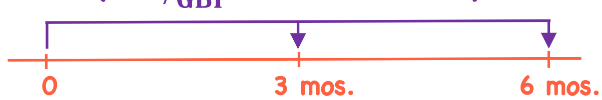
⇒ value of contract at initiation = zero
⇒ MTM value changes as:
spot changes
interest rates change (foreign or domestic)

e.g. $(^{AUD}/_{GBP}$ 1.6210/1.6215) price base



bought £10M for $^{AUD}/_{GBP}$ = 1.6100
(i.e. will pay 16.1M AUD in 6 months)

wants to close position
- must sell £10M 3 mos. forward at a spot today (3 mos. points 130/140)

1. Determine bid 1.6210 + 130 = 1.6340
2. Calculate value in AUD (1.6340 − 1.6100) × 10M
= .0240 × 10M = 240,000 AUD

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3. Discount to present using the 3-mos. Libor rate

3 mos. 240,000 AUD

AUD 3-mos. Libor = 4.8%

(discount rate = currency)

$$\frac{240{,}000}{[1+.048(^{90}/_{360})]} = 237{,}154 \text{ AUD}$$

e.g./ • 6-mos. ago dealer sold 1 million CHF against the GBP, 180-day term
• all-in rate of 1.4850 $(^{CHF}/_{GBP})$
• dealer now wants to roll position forward another 6-mos.

current spot + points

Spot rate (CHF/GBP)	1.4939/1.4941
One month	-8.3/-7.9
Two month	-17.4/-16.8
Three month	-25.4/-24.6
Four month	-35.4/-34.2
Five month	-45.9/-44.1
Six month	-56.5/-54.0

• Dealer will use an FX swap
- simultaneous spot & forward transactions
- buy 1M CHF spot, sell 1M CHF forward
- common spot rate applied to both legs ⇒ mid-market of current spot

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current spot + points

Spot rate (CHF/GBP)	1.4939/1.4941
One month	-8.3/-7.9
Two month	-17.4/-16.8
Three month	-25.4/-24.6
Four month	-35.4/-34.2
Five month	-45.9/-44.1
Six month	-56.5/-54.0

original all-in = 1.4850

1. current all-in rate for delivery of GBP against CHF in 3 months?

what are we doing with the base? selling

Selling happens on the bid! So… 1.4939 − 25.4 = 1.49136

2. CF dealer realizes on settlement?

Day 1 - Sold CHF, long GBP $\frac{1,000,000}{1.4850} = 673,400.67$

Settlement - Buy CHF, Sell GBP $\frac{-1,000,000}{1.4940} = \frac{-669,344.04}{£\ 4056.63}$ inflow

3. All-in rate to sell CHF against GBP 6-mos. (another dealer)

what are we doing with the base? buying

1.4941 − 54 = 1.4886

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e.g./ • sold 10 million NZD against USD
• all-in forward price = 0.7900 $\left({}^{USD}/_{NZD}\right)$
• 3 mos. to expiration, mark-to-market the position

Spot ${}^{USD}/_{NZD}$ 0.7825/0.7830
3-mos. points -12.1/-10.0
3-mos. Libor NZD = 3.31%
3-mos. Libor USD = 0.31%

① Buy 10 million NZD against USD
3 mos. forward 0.7830
$$\begin{array}{r} 0.7830 \\ -\quad 10 \\ \hline 0.7820 \end{array}$$

② (0.7900 - 0.7820) × 10M = 80,000 USD

③ $\left[\dfrac{80{,}000}{1+.0031\left({}^{90}/_{360}\right)}\right] = 79{,}938.05$

e.g. 2/ reverse the position above (i.e. sold \$10M USD against NZD)

MTM?

① Sell NZD
.7825 - .00121 = .78129

② $\left[\left(\dfrac{1}{.7900}\right)-\left(\dfrac{1}{.78129}\right)\right] \times 10\text{M}$
= -141,117 NZD

③ Disc. by 3.31%
$\left[\dfrac{-141{,}117}{1+.0331\left({}^{90}/_{360}\right)}\right] = (-139{,}959)$

Parity Relations

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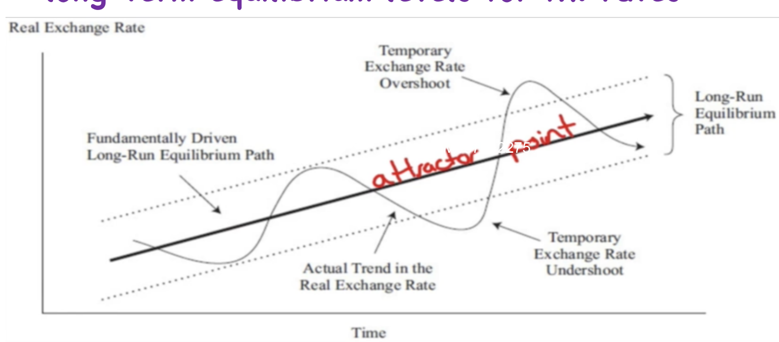
⇒ a framework for developing a view about future exchange rate movements

- parity conditions → describe the inter-relationships that jointly determine long-run movements in fx., interest rates & inflation
- are the basic building blocks for describing long-term equilibrium levels for fx. rates



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- Parity conditions show how:
  - expected inflation differentials
  - interest rate differentials
  - forward fx. rates
  - current spot fx. rates
  - expected future spot fx. rates

would be linked in an ideal world

1) Covered Interest Rate Parity
- describes a riskless, arbitrage relationship
- supported where capital is permitted to flow freely, spot & forward fx. markets are liquid, financial market conditions are basically stress-free

empirical studies
- international parity conditions rarely hold in either the short or medium term (except covered interest rate parity)
- further, forward fx. rates have been found to be poor predictors of future spot fx. rates

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2) Uncovered Interest Rate Parity/ expected return on an uncovered (unhedged) foreign currency investment should equal the return on a comparable domestic currency investment

e.g./ $(1 + i_d) = 1.05$ or 5% return $\quad$ $(1 + i_f) = 1.10$ or 10% return

UIRP implies $(1 + i_f)(1 - \%\Delta S_{f/d}) - 1 = i_d$

simplifies to: $i_f - \%\Delta S^{e}_{f/d} = i_d$ (e = expected)

or/ $\underbrace{i_f - i_d}_{\text{interest rate differential}} = \underbrace{\%\Delta S^{e}_{f/d}}_{\text{expected \%age change in spot fx.}}$

Note: even when UIRP holds, S.d. of returns on i_d = 0, s.d. of returns on $i_f - \%\Delta S_{f/d} \neq 0$

∴ UIRP assumes risk-neutral investors

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2) Uncovered Interest Rate Parity/ - under UIRP country with higher interest rate is expected to see the value of its currency depreciate

- if UIRP held
  - no excess return could be gained by going long a high-yielding currency & short a low-yielding one
  - no incentive to shift capital from one currency to another

- empirical studies ⇒ fails to hold over short & medium terms
  - interest rate differentials are generally a poor predictor of future fx. rate changes
  - high-yield currencies tend to strengthen, not weaken
  - forward fx. rates are a poor predictor of future spot fx. rates
  - finally, current spot fx. rates are also poor predictors of future spot fx. rates

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2. Uncovered Interest Rate Parity/

e.g./ $^{JPY}/_{AUD}$ spot mid market 79.25

base = domestic

one-year forward points	-301.9
one-year AUD deposit rate	5%
one-year JPY deposit rate	1%

forecasting that the $^{JPY}/_{AUD}$ spot 1 yr. from now = 79.25
assumes: spot rates follow a random walk

1. Based on UIRP, over next year, $S^e_{f/d}$? ($i_d - i_f$ = .05 - .01 = .04)

domestic gets 5%, foreign gets 1%, so foreign would also have to buy 4% more domestic currency ∴ JPY ↑ or AUD ↓ by 4%

2. May not be accurate prediction because: no arb. condition forces UIRP to hold

3. Using forward points to predict future JPY/AUD spot rate one year ahead assumes: UIRP holds ⟶ risk neutrality

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3) Purchasing Power Parity/ the relationship
(PPP) between fx. rates and inflation differentials

- based on 'law of one price' ⇒ identical goods should trade at the same price across countries when valued in terms of a common currency

• foreign price of a good P_f^x • domestic price of a good P_d^x

$P_f^x = S_{f/d} \bullet P_d^x$ if $P_{EUR}^x = €100$ & $S_{USD/EUR} = 1.4000$, then $P_{USD}^x = \$140$

- absolute version of PPP - extends to the general price level

$$\underbrace{P_f = S_{f/d} \bullet P_d}_{S_{f/d} = P_f / P_d}$$

- assumes that all domestic & foreign goods are tradable & that price indices (d & f) are identical (same g & s, same weights)

- equilibrium fx. rate between 2 countries determined by the ratio of their nominal price levels

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3) Purchasing Power Parity/ - highly unlikely to hold
(PPP) - assumes 'goods arbitrage' will equate prices

$\underbrace{\text{(transaction costs, trade impediments)}}$

⇒ relative version of PPP ⟶ assumes these costs are constant over time

then $\%\Delta S_{f/d} \cong \pi_f - \pi_d$ π - inflation rate

changes in fx. rates inflation differential

- basically, the currency of the high inflation country should depreciate relative to the currency of the low inflation country

⇒ ex ante version of PPP $\%\Delta S_{f/d}^e = \pi_f^e - \pi_d^e$ - expected changes in spot fx. rates are driven by expected differences in inflation rates

- countries that are expected to run persistently high inflation rates should expect to see their currencies dep. over time.

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3) Purchasing Power Parity/ - inflation differentials
(PPP) across countries cause nominal fx. rates to adjust in order to re-equilibrate real purchasing power

$\underbrace{\text{adjust in order to re-equilibrate real purchasing power}}$

concept of real fx. rates ⇒ captures real purchasing power of a currency, defined in terms of the amount of real g/s it can purchase internationally

$$\underbrace{q_{f/d}}_{\text{real fx. rate}} = \frac{\overbrace{S_{f/d}P_d}^{\text{domestic price level in foreign currency}}}{P_f} = S_{f/d}\left(\frac{P_d}{P_f}\right) = S_{f/d}\left(\frac{CPI_d}{CPI_f}\right)$$

⇒ empirical studies: real fx. rates 'mean revert' in the long-run i.e. nominal fx. rates gradually gravitate towards their PPP-based values (long-run PPP equilibrium values)

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4) The Fisher effect & Real Interest Rate Parity/
- combines both
• interest rate differentials
• inflation differentials

$i = r + \pi^e$ - nominal rate = real rate + expected inflation

So... $i_d = r_d + \pi_d^e$ ⇒ $i_f - i_d = \underbrace{(r_f - r_d)}_{\text{yield spread}} + \underbrace{(\pi_f^e - \pi_d^e)}_{\text{inflation differential}}$

$i_f = r_f + \pi_f^e$

- since we observe i_f & i_d , and not r_f & r_d

$(r_f - r_d) = (i_f - i_d) - (\pi_f^e - \pi_d^e)$

from UIRP
$i_f - i_d = \%\Delta S_{f/d}^e$

from PPP
$(\pi_f^e - \pi_d^e) = \%\Delta S_{f/d}^e$

substituting: $\underbrace{(r_f - r_d) = \%\Delta S_{f/d}^e - \%\Delta S_{f/d}^e = 0}$

if both UIRP & ex ante PPP hold, the real yield spread $(r_f - r_d) = 0$ ⇒ real interest rate parity

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4) The Fisher effect & Real Interest Rate Parity/

$\mathrm{r_f - r_d = 0}$ implies $\underbrace{\mathrm{i_f - i_d = \pi_f^e - \pi_d^e}}_{\text{the International Fisher effect}}$

- empirical studies/ - there exists an interaction between nominal interest rate, exchange rates, and inflation rates across countries (over longer time periods)

∴ international parity conditions serve as an anchor (or attractor point) for longer-term exchange rate movements

Dr. Mark goes through some questions in the video

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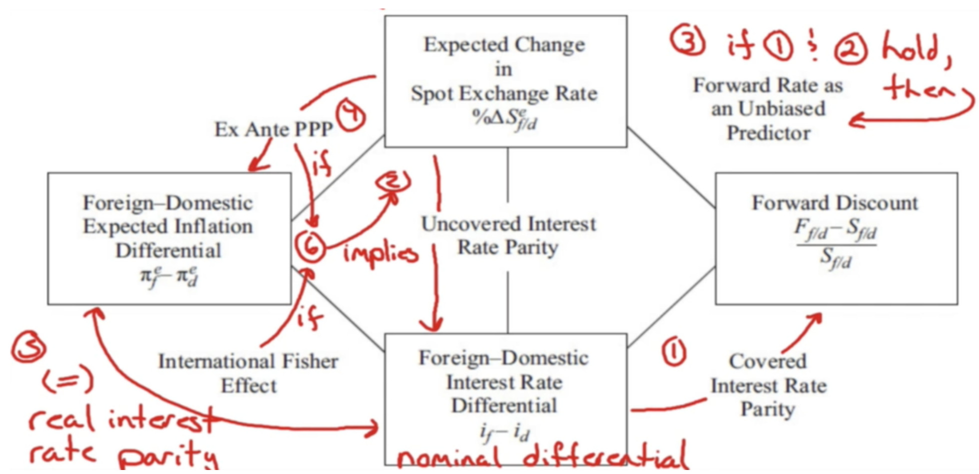
- if all International parity conditions held:

$\%\Delta S^e_{f/d} =$ forward premium/discount (%'age) $= \mathrm{i_f - i_d = \pi_f^e - \pi_d^e}$



- if all conditions held, it would be impossible for a global investor to earn consistent profits on currency movements

Dr. Mark goes through some questions in the video

Assessing Equilibrium

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⇒ IMF Consultative Group on Exchange Rate Issues

3 pronged approach to derive long-run equilibrium exchange rate assessments

1. Macroeconomic balance approach – estimates how much fx. rates need to adjust in order to close the gap between the medium-term expectation for a country's current account imbalance and that country's normal (or sustainable) current account imbalance (focus on 'real' flows)

2. External sustainability approach ⇒ focuses on stocks of outstanding assets or debt ⇒ how much fx. rates need to adjust to ensure that a country's net foreign asset/GDP ratio or net foreign liability/GDP ratio stabilizes at some benchmark level (focus on 'financial' flows)

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⇒ IMF Consultative Group on Exchange Rate Issues

3. a reduced form econometric model – seeks to estimate the equilibrium path that a currency should take on the basis of the trends in several key macro variables
(e.g. net foreign asset position, terms of trade, relative productivity)

⇒ empirical observations/
- have some predictive power regarding future changes in real exchange rates
- predict direction but poor on magnitude
- also tend to overpredict

- International parity relations specify relationships, these 3 approaches attempt to explain some of the <u>mechanisms</u> that help realign fx. rates

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Unsustainably Large:	PPP Overvaluation	PPP Undervaluation
- Current Acct. Surplus	1	3
- Current Acct. Deficit	2	4

	action required to bring fx.-rate closer to long-run fair value
1. overvalued currency, trade surplus	unclear
2. overvalued currency, trade deficit	encourage depreciation
3. undervalued currency, trade surplus	encourage appreciation
4. undervalued currency, trade deficit	unclear

Carry Trade

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UIRP → high-yield currencies are expected to depreciate
UIRP → low-yield currencies are expected to appreciate

significant evidence that UIRP does not hold over short and medium term time periods

opens up a trading strategy called the 'FX Carry Trade'

- take long positions in high yield currencies and short positions in low yield currencies (funding currencies)
- earns positive excess returns in most market conditions (low volatility)
- but, volatility/perceived risk can cause these positions to unwind fast (crash risk)

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⇒ carry trades use considerable leverage
(50x – 200x)

⇒ use of a volatility filter – if avg. level fx. volatility falls below some threshold, open carry trade positions (may be accompanied by some other vol. measure)

- if above, close carry trade positions

→ use of PPP

avoid

↑ risk of depreciation — upper PPP band

long high yield curr.

short low yield curr.

↓ risk of appreciation — lower PPP band

- Policy Perspective – concern that carry trade activities may be generating fx. rate misalignments
  - unwinding a large carry trade could cause a financial crisis

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	1-yr. Libor	Pair	1-yr. later
JPY	0.10%	JPY/USD → 81.30	80.00
AUD	4.50%	USD/AUD → 1.0750	1.0803

- borrow in yen, invest in 1-yr. AUD Libor

- after 1 yr., calculate the all-in return to this trade

$$JPY/AUD = JPY/USD \times USD/AUD$$

original
81.30 × 1.0750 = 87.40

1-yr. later
80.00 × 1.0803 = 86.42

$$\text{All-in} = \frac{86.42}{87.40}(1.0450) - .001$$

$$= 1.033283 - .001$$

$$= 1.032283 \sim 3.23\%$$

Balance of Payments Impacts

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⇒ Current Acct.: sum of all recorded transactions in traded good, services, income & net transfer payments

- persistent curr. act. deficits ⇒ currency depreciation over time
(surplus) (appreciation)

$\underbrace{|\text{Capital Acct.}|}_{\text{financial flows}} = |\text{Current Acct.}|$

financial flows ⇒ investment/financing decisions are usually the dominant factor in determining exchange rate movements (short to intermediate term)

Why?/ ① prices of real goods/services tend to adjust much more slowly - financial assets adjust quickly, leading to inflows & outflows ⇒ impact the fx. rate

Page 33
LOS j
- explain

Why?/ ② Production of real g/s takes time (demand decisions subject to substantial inertia)
- liquid financial markets instantaneous redirection of financial flows

③ current spending/production decisions reflect only purchases/sales of current production
- investment/financing decisions reflect not only the financing of current expenditures but also reallocation of existing portfolios

④ expected exchange rate movements can induce very large short-term capital flows

∴ actual fx. rate very sensitive to the currency views held by owners/managers of liquid assets

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- explain

⇒ Current Account Imbalances/

1) the flow supply/demand channel

- purchases/sales of internationally traded g/s require the exchange of domestic & foreign currencies

trade surplus → more demand for currency → upward pressure

trade deficit → more selling of currency → downward pressure

- persistent surpluses/deficits should revert

- surplus country currency would appreciate to the point that exports would drop, imports would increase

- amount of adjustment (restore current acts. to balanced position) depends on:

a) Initial gap between X & M ⇒ substantial fx. depreciation if:

i) import demand more price elastic (vs. export demand)

ii) the wider the deficit gap

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- explain

⇒ Current Account Imbalances/

1) the flow supply/demand channel

- amount of adjustment (restore current acts. to balanced position) depends on:

b) response of X & M prices to Δ(fx. rate)

- typically, fx. dep. ⇒ increase in import prices

⇒ decreases in export prices

empirical results - limited pass-through effects of Δ(fx. rate)

c) response of X & M demand to Δ(X & M prices)

- empirical results ⇒ demand quite sluggish

2) the portfolio balance channel - surplus nations will build up fx. reserves → rebalancing these reserves result in selling foreign currencies & buying domestic currencies

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- explain

⇒ Current Account Imbalances/

3) the debt sustainability channel

- persistent trade deficits will result in increasing debt owed to foreigners

- if levels become unsustainable, major depreciation of currency will be required (rebalance current acct.)

⇒ Capital Account (capital flows)

- greater global capital market integration has resulted in increased freedom of capital to flow across national borders

- can have negative consequences (hot money)

- introduction of capital controls

- attempts to prevent capital inflows from pushing currency values to overvalued levels

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- explain

⇒ Capital Account (capital flows)

a) Real Interest Rate Differentials, Capital Flows, & the FX Rate/

$$\underbrace{q_{f/d}}_{\text{real fx. rate}} = \underbrace{\bar{q}_{f/d}}_{\text{real long-run equilibrium rate}} + [\underbrace{(r_d - r_f)}_{\text{real interest rate differential}} - \underbrace{(\varphi_d - \varphi_f)}_{\text{relative risk premia}}]$$

real interest rate differential (play a major role)

relative risk premia → perceived sustainability of the country's external balances

⇒ in short, capital responds to interest rate differentials, risk premia & expectations w.r.t. fx. rate movements

b) Interest Rate Differentials, Carry Trades, & FX Rates/

$$q_{L/H} = \bar{q}_{L/H} + \underbrace{(i_H - i_L)}_{\text{yield spread widens}} - \underbrace{(\pi^e_H - \pi^e_L)}_{\text{inflation expectations spread tightens}} - \underbrace{(\varphi_H - \varphi_L)}_{\text{risk premia spread tightens}}$$

will rise when

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- explain

⇒ Capital Account (capital flows)

b) Interest Rate Differentials, Carry Trades, & FX Rates/

e.g./ high yield, high inflation rate country ⇒ wants price stability + long-term sustainable growth

Note: unobservable

$$q_{L/H} = \bar{q}_{L/H} + \underbrace{(i_H - i_L)}_{\text{widen}} - \underbrace{(\pi^e_H - \pi^e_L)}_{\text{tighten}} - \underbrace{(\varphi_H - \varphi_L)}_{\text{tighten}}$$

(1 + 2) upward revised assessment of the long-run equilibrium value

1) combat inflation by raising interest rates (monetary)

2) policies of lower gov't. debt, financial market liberalization, removal of capital flow restrictions, attraction of FDI (economic & fiscal)

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- explain

⇒ Capital Account (capital flows)

b) Interest Rate Differentials, Carry Trades, & FX Rates/

- combination of monetary, fiscal, & economic policies contribute to continued upward pressure on fx. rate

∴ Carry Trade profitability = $\underbrace{f(\text{monetary, fiscal, economic policies})}_{\text{proceed gradually, reinforce each other, thus produce persistence in carry trade returns}}$

c) Equity Market Trends & FX Rates/

- not a stable relationship - instability in correlation

- makes it difficult to form judgments on possible future fx. moves based solely on expected equity market performance

Monetary & Fiscal Policies

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- explain

⇒ Mundell-Fleming Model/

Expansionary (increase AD & output)

Monetary Policy

- reduce interest rates

⇒ increase investment & consumption spending

- leads to capital outflows
  - downward pressure on fx. rates

Fiscal Policy

- increase spending/lower taxes
- exerts upward pressure on interest rates (larger deficits need to be financed)
- leads to capital inflows
  - upward pressure on fx. rates

- the more responsive capital flows are to interest rate differentials:
  - larger the fx. depreciation
  - larger the fx. appreciation

$\underbrace{\text{depreciation}}_{\text{eventually boosts net exports}}$

eventually boosts net exports

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- explain

⇒ Mundell-Fleming Model/

w/ fixed fx. rates

Expansionary (increase AD & output)

Monetary Policy

- central bank will have to buy its own currency & deplete fx. reserves (to support currency)

(limited by stock of reserves)

- this tightens domestic credit conditions ⇒ offsets intended expansionary policy

Fiscal Policy

- central bank will have to sell its currency and build reserves

(theoretically unlimited)

- this expands domestic money supply ⇒ reinforces expansionary policy

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- explain

⇒ Mundell-Fleming Model/

Key points/ ① if domestic policymakers try to
a) pursue independent monetary policy
b) permit capital to flow freely across national borders
c) defend fx. rates

all 3 cannot be satisfied at same time

② capital controls → b)
- may need to be imposed to satisfy a) & c)

high capital mobility

	Expansionary Monetary Policy	Restrictive Monetary Policy
Expansionary Fiscal Policy	Ambiguous	Domestic currency appreciates
Restrictive Fiscal Policy	Domestic currency depreciates	Ambiguous

low capital mobility - effects primarily through trade flows

	Expansionary Monetary Policy	Restrictive Monetary Policy
Expansionary Fiscal Policy	trade balance Domestic currency depreciates depreciates	Ambiguous
Restrictive Fiscal Policy	Ambiguous	trade balance Domestic currency appreciates improves

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- explain

⇒ The Monetary Approach (with flexible prices)

Mundell/Fleming ⇒ monetary policy → interest rates → fx. rates

monetary approach ⇒ monetary policy → price level/rate of inflation → fx. rates

as per quantity theory of money ⇒ money supply changes are the primary determinant of price level changes
- assuming PPP holds ⇒ a money supply induced increase in domestic price levels relative to foreign prices should lead to a proportional decline in the domestic currency's value
- shortcoming: assumption of PPP holding at all times
∴ monetary approach may not realistically explain policy effects on the fx. rate

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- explain

⇒ The Monetary Approach (with somewhat flexible prices)

- Dornbusch Overshooting Model

- prices exhibit limited flexibility in $\underbrace{\text{the short run,}}_{\uparrow M\text{ , CPI unch.}}$ $\underbrace{\text{fully flexible in the long run}}_{\uparrow M,\ \uparrow CPI\ \Rightarrow\ \downarrow FX\text{ rate}}$

↑ real M, ↓ i_d ⇒ capital outflow,

↓ fx. rate but overshoots on a short run basis

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- explain

⇒ Fiscal Policy & FX Rates/

Mundell Fleming ⇒ short run model of fx. rate determination

- makes no allowance for the long-term effects of budgetary imbalances that typically arise from fiscal policy actions

⇒ portfolio balance approach ⇒ fixes this

- global investors hold diversified portfolios of domestic & foreign assets

- a steady increase in the supply of domestic bonds outstanding (widening deficit/debt) will be willingly held only if holders are compensated with higher expected return

1) higher interest rates

2) depreciation of the currency to a level sufficient to generate anticipation of gains from subsequent appreciation

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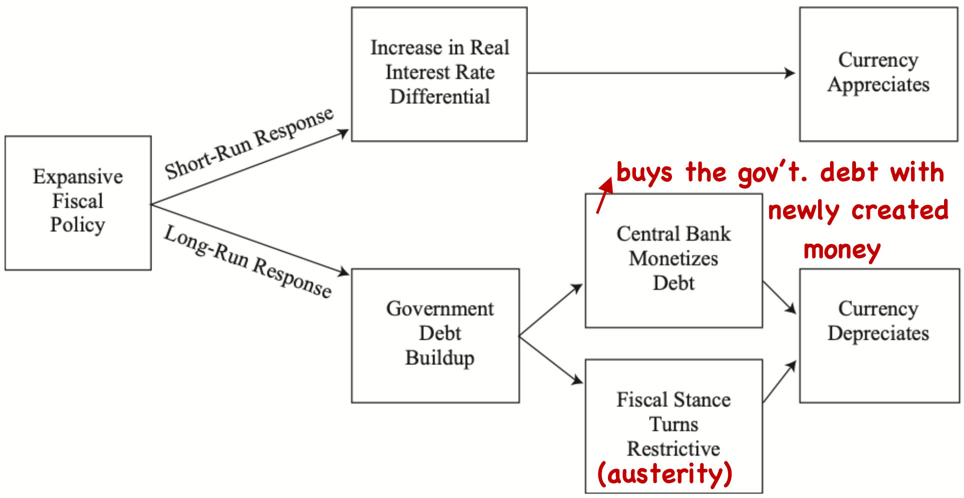
- explain

⇒ Fiscal Policy & FX Rates/

- combining MF & portfolio balance approach:



Intervention

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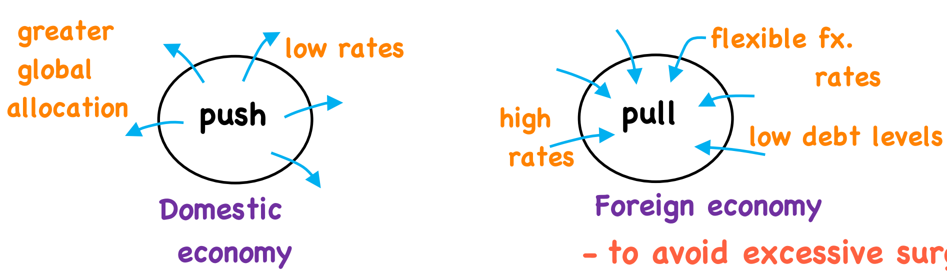
- describe



- to avoid excessive surges of capital inflows (typically followed by huge outflows) use of 1) capital controls
2) fx. market intervention

⇒ as per IMF guidelines/

1) undervalued currency - no intervention

2) fairly or overvalued currency, no inflation threat
- unsterilized intervention - expand M, will cause rates to move lower, slow down capital inflows

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- describe

3) fairly or overvalued currency, inflation is a threat
⇒ sterilized intervention
- sell domestic securities to the private sector
⇒ mops up excess liquidity created by the FX intervention

4) if there is a limit to sterilization → tighten fiscal policy to slow down domestic demand

⇒ if intervention fails ⇒ capital controls

Effectiveness/ - developed economies ⇒ effect statistically insignificant
- emerging economies ⇒ contributes to lower EM fx. volatility but no stat. sig. effect on level
- but greater ability to influence level & path than dev. countries
- capital controls → mixed
- the more persistent the inflows, the larger their magnitude, less likely capital controls will work

Currency Crises - Warning Signs

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- describe

1. $q_{f/d}$ substantially higher than its mean level during tranquil periods
2. FX reserves decline precipitously as the crisis approaches
3. Some deterioration in terms of trade $\left(X_P/M_P\right)$
4. CPI significantly higher pre-crisis vs. tranquil periods
5. M2/bank reserves ⇒ tends to rise in the 24-month period leading up to a crisis
6. Broad money growth (nominal & real) rises sharply in the 2 yrs. leading up to a crisis
7. Nominal private credit growth rises sharply pre-crisis
8. Typically some form of equity market boom

Economic Growth

a. compare factors favoring and limiting economic growth in developed and developing economies

b. describe the relation between the long-run rate of stock market appreciation and the sustainable growth rate of the economy

c. explain why potential GDP and its growth rate matter for equity and fixed income investors

d. contrast capital deepening investment and technological progress and explain how each affects economic growth and labor productivity

e. demonstrate forecasting potential GDP based on growth accounting relations

f. explain how natural resources affect economic growth and evaluate the argument that limited availability of natural resources constrains economic growth

g. explain how demographics, immigration, and labor force participation affect the rate and sustainability of economic growth

h. explain how investment in physical capital, human capital, and technological development affects economic growth

i. compare classical growth theory, neoclassical growth theory, and endogenous growth theory

j. explain and evaluate convergence hypotheses

k. describe the economic rationale for governments to provide incentives to private investment in technology and knowledge

l. describe the expected impact of removing trade barriers on capital investment and profits, employment and wages, and growth in the economies involved

Growth Factors

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- compare

Factors → Economic Growth → earnings expectations → equity prices

Economic Growth → growth rate of real incomes → real interest rate → real returns

Economic Growth → (actual-potential) → inflation expectations → bond prices

Factors → identify & forecast

$\underbrace{\text{(actual-potential)}}$ → long-run aggregate output
- growth rate of potential GDP = economy's sustainable growth rate
- max. amt. an economy can sustainably produce without inducing an increase in the growth rate of inflation

Page 2
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- compare

- comparing performance/

$\%\Delta GDP_r$ - annual %age change in real GDP
(measures how rapidly the economy is expanding)

$\%\Delta GDP_r/capita$ - annual %age change in real GDP/capita
(reflects the average $\underbrace{\text{standard of living}}_{\text{material well-being}}$)

- to compare: convert each country's real GDP to a common currency (typically USD) at a $\underbrace{\text{PPP exchange rate}}_{\text{why?}}$

Why not market fx. rates?
1) fx. rates too volatile
2) fx. rates reflect traded g/s + financial flows

(much of GDP is non-traded)

assume/

Country A	haircut = \$20
Country B	haircut = \$5

but same material well-being
∴ should be same value in GDP

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LOS a
- compare

- comparing performance/
  - developed (high GDP/capita) vs. developing (low GDP/capita)
  - explanations (of the diversity)/(beyond economic factors such as Labour, Kapital, Tech.)

1. Savings & Investment

low S ⇒ low I ⇒ Slow GDP growth ⇒ persistently low incomes

remedies typically aimed here
- capital flows
- FDI

2. Financial Markets & Intermediaries/

• promote growth by ① allocating capital to projects with the highest risk-adjusted returns

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LOS a
- compare

2. Financial Markets & Intermediaries/

- promote growth by
  ② encouraging savings by creating investment products
  ③ overcoming credit constraints (intermediation)

- countries with better-functioning financial markets & intermediaries grow at a faster rate

3. Political Stability, Rule of Law, and Property Rights

- Political Stability: Stable & effective government
- Rule of Law: • well-developed legal/regulatory system
- Property Rights: • protection of private property, including intellectual property

4. Education & Health Care Systems/

Education: raises skill level (if they stay)
- contributes to potential GDP

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- compare

4. Education & Health Care Systems/ ⇒ speaks to labour availability & willingness to invest

- also raises the productivity of existing capital

(e.g. formal schooling, on-the-job training)

Focus of Innovation

leading edge
- investments in post secondary

most effective

imitation
- investments in primary & secondary

5. Tax & Regulatory Systems

limited regulation

6. Free Trade & Unrestricted Capital Flows

- more goods at lower cost
- access to larger markets for domestic firms

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- compare

6. Free Trade & Unrestricted Capital Flows

- foreign companies can invest directly (FDI)
    - building/buying property/plant/equipment
- foreign investors can purchase securities issued by domestic countries

both ↑ capital stock → ↑ productivity employment, wages → ↑ domestic savings

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	Real GDP Per Capita in Dollars				
	Venezuela	**Argentina**	**Singapore**	**Japan**	**South Korea**
1950	$8,104	$6,164	$4,299	$3,048	$1,185
2010	$10,560	$13,468	$56,224	$34,828	$30,079

growth rates $\left(\frac{10{,}560}{8{,}104}\right)^{\frac{1}{60}}-1$ $\left(\frac{13{,}468}{6{,}164}\right)^{\frac{1}{60}}-1$ $\left(\frac{56{,}224}{4{,}229}\right)^{\frac{1}{60}}-1$ $\left(\frac{34{,}828}{3{,}048}\right)^{\frac{1}{60}}-1$ $\left(\frac{30{,}079}{1185}\right)^{\frac{1}{60}}-1$

= 0.44% = 1.31% = 4.38% = 4.14% = 5.54%

rank (5) (4) (2) (3) (1)

growth needed to equal SK GDP/capita $\left(\frac{30079}{8104}\right)^{\frac{1}{60}}-1$ = 2.21% $\left(\frac{30079}{6164}\right)^{\frac{1}{60}}-1$ = 2.68%

- small differences in growth rates matter over long periods if time

Venezuela
- plans to invest in infrastructure, education, health care
- no legal reforms, no strengthening of property rights, foreign Inv. discouraged

(Thoughts?)

Stock Market vs. Economy

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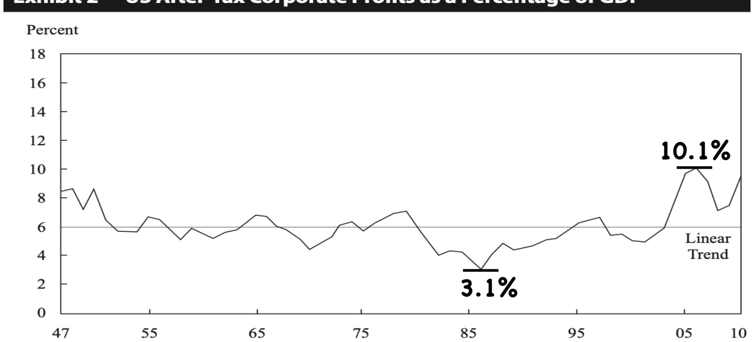
- potential GDP ⇒ places a limit on how fast an economy can grow

Q: is earnings growth also limited by growth rate of potential GDP?

for earnings growth > potential GDP growth

ratio of $\frac{\text{Corp. profits}}{\text{GDP}}$ must be increasing (limits out)

Exhibit 2 US After-Tax Corporate Profits as a Percentage of GDP



Source: FRED database, Federal Reserve Bank of St. Louis.

- in the long-run, earnings growth cannot exceed the growth rate of potential GDP

Stationary relationship

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- describe
- explain

⇒ Forecasting Equity Returns/

Discounted Cash Flow Approach/

$P_0 = \frac{D_0(1+g)}{r-g}$ ⇒ solved for r: $r = \frac{D_1}{P_0} + g$

$\frac{D_1}{P_0}$ – div. yield (income)

g – cap. appreciation (growth) (long term g = trend GDP growth)

Consider:		s.d.	
	S&P500	16.1	volatility of mean return in historical estimates
	P/E	28.5	
	E/GDP	28.9	
	rGDP	3.0	g ⇒ much more stable (advantage of GGM approach)
	π	3.2	

⇒ Grinold-Kroner model

$$E(R_e) = \underbrace{\left({}^{D}/_{P} - \%\Delta S\right)}_{\substack{\text{expected cash}\\ \text{flow return}\\ \text{(income)}}} + \underbrace{\underset{\substack{\text{expected}\\ \text{nominal earnings}\\ \text{growth}}}{\%\Delta E} + \underset{\substack{\text{expected repricing}\\ \text{return}}}{\%\Delta\, {}^{P}/_{E}}}_{\text{growth}}$$

where:
- ${}^{D}/_{P}$ – dividend yield
- E – total earnings
- S – shares outstanding
- ${}^{P}/_{E}$ – price-earnings ratio

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- describe
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⇒ Fixed Income/

growth rate of potential GDP (real incomes rise) determinant of:
- level of real interest rates
- ∴ real asset returns in general

$(y - y^*)$ → as $y^*\uparrow$, removes inflationary pressure even as y increases, thus higher incomes require higher rates to induce savings

- also/

1. higher g in y^* improves the general credit quality of issuers
   (sovereign most directly, corporate as well)
2. $(y - y^*)$ important w.r.t. Central bank policy changes
   (Taylor Rule)
3. $\left({}^{debt}/_{y} \text{ vs. } {}^{debt}/_{y^*}\right)$ – structural or cyclical adjusted debt

Determinants of Economic Growth

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LOS d
- distinguish
- explain

⇒ Simple Production Function/

$Y = AF(K,L)$

aggregate output

TFP – a scale factor referred to as Total Factor Productivity

• inputs called factors of production
- capital (K)
- labor (L)

(represents another input
- technology/level of productivity)

- Cobb-Douglas prod. function/

$F(K,L) = K^{\alpha}L^{1-\alpha}$ $(0 < \alpha < 1)$

- embodies cumulative effects of scientific advances, applied R&D, improvements in mgmt. methods & ways of organizing

- factors of production are paid their marginal product
- profit max requires:

MPK = rental price of capital (r)

MPL = real wage rate

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- distinguish
- explain

⇒ Simple Production Function/

$Y = AF(K,L)$ where $F(\ ) = K^{\alpha}L^{1-\alpha}$

$$\frac{dY}{dK} = MPK = \alpha AK^{\alpha-1}L^{1-\alpha}$$

$$= \alpha\frac{(AK^{\alpha}L^{1-\alpha})}{K} = \frac{\alpha Y}{K} = r$$

$rK = \alpha Y$

$\alpha = \frac{rK}{Y}$ – capital income

- in words, α is the share of GDP paid out to the suppliers of capital

$$\frac{dY}{dL} = MPL = (1-\alpha)AK^{\alpha}L^{-\alpha}$$

$$= (1-\alpha)\left(AK^{\alpha}\frac{L^{1-\alpha}}{L}\right)$$

$$= \frac{(1-\alpha)Y}{L}$$ = real wage rate (w_r)

$Lw_r = (1-\alpha)Y$

$(1-\alpha) = \frac{Lw_r}{Y}$

share of GDP paid out to suppliers of labour

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- distinguish
- explain

⇒ Simple Production Function/

Cobb-Douglas ⇒ 2 properties/

① Constant returns to scale

double inputs ⇒ double outputs

$Y = AF(K,L)$ → multiplying by $^1/_L$ → output per worker, or labor productivity

$$\underbrace{\overset{y}{^Y/_L} = AF(^K/_L, ^L/_L)}_{y = AF(k,1) = Ak^{\alpha}} = AF(\underbrace{^K/_L}_{\text{capital-to-labor ratio } (k)}, 1)$$

$$\left[^Y/_L = A\left(\overbrace{^K/_L}\right)^{\alpha}\left(^L/_L\right)^{1-\alpha}\right]$$

- the amount of goods a worker can produce depends on the amount of capital for each worker $(^K/_L)$, TFP (A) & the share of capital in GDP (α)

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- distinguish
- explain

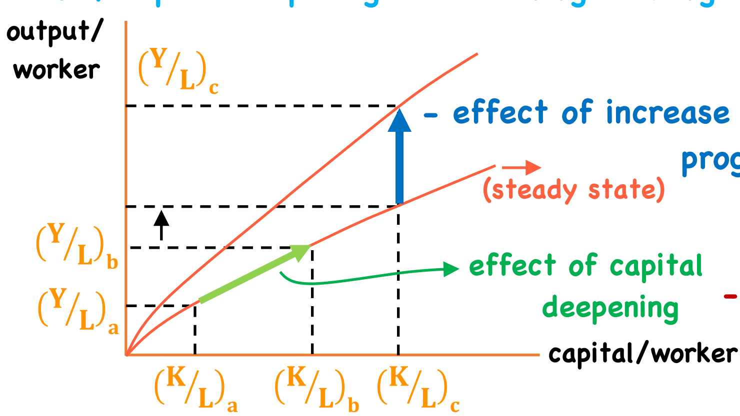
⇒ Simple Production Function/

Cobb-Douglas ⇒ 2 properties/

② diminishing marginal productivity - the extra output obtained from each additional unit of input will decline

α close to zero - diminishing marginal returns to capital very significant

now/ Capital Deepening vs. Technological Progress/



- effect of increase in technological progress (TFP)

(higher output at each level of K)

- an ↑ in $(^K/_L)$ - reflects rising investment
- stops when MPK = r

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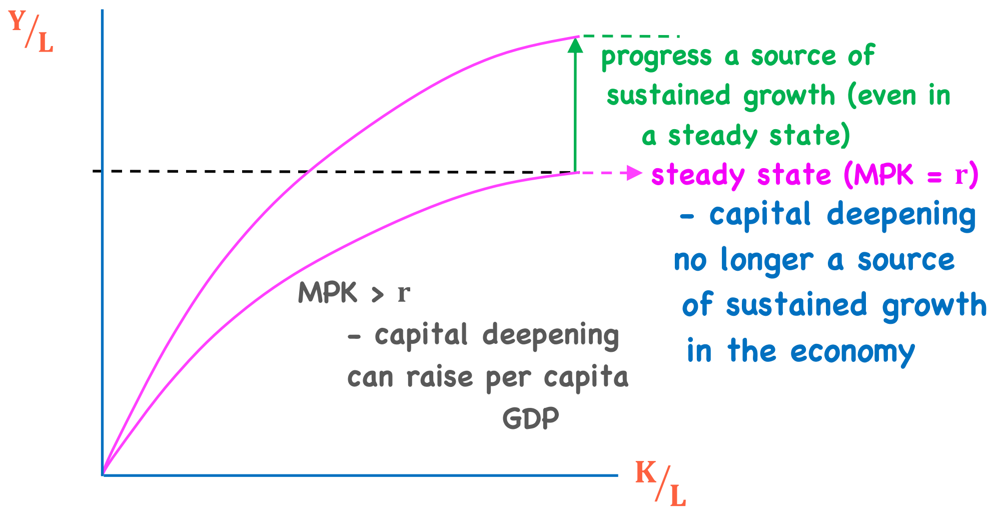
LOS d
- distinguish
- explain

⇒ Simple Production Function/



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- distinguish
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e.g./ Country A $\left(K/L\right)$ = \$100,000

Country B $\left(K/L\right)$ = \$5,000

- what effect on growth rate of potential GDP of:

① ↑ I in both countries

- capital deepening, ↑ $\left(K/L\right)$
  - A - small effect
  - B - large effect

② ↑ university research in both countries

- likely impact on TFP
  - A - raise potential GDP
  - B - raise potential GDP

③ eliminate restrictions on foreign investment for Country B

I ↑ - raises potential GDP in B

Growth Accounting

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– forecast
– explain

– growth accounting equation/ (used to measure potential GDP)

$$\underbrace{\frac{\Delta Y}{Y}}_{\text{growth rate of output}} = \underbrace{\frac{\Delta A}{A}}_{\text{rate of technological change}} + \alpha \underbrace{\frac{\Delta K}{K}}_{\text{growth rate of capital}} + (1-\alpha)\underbrace{\frac{\Delta L}{L}}_{\text{growth rate of labour}}$$

α – elasticity of output w.r.t. capital

$(1-\alpha)$ – elasticity of output w.r.t. labour

estimated from GDP accounts

$(\alpha \sim .3 \text{ for U.S.})$

∴ increase in growth rate of labour will have a larger impact than capital

– Y, L & K data are available from GDP accounts

– A must be estimated – a sort of residual
  – the amount of growth in output that cannot be explained by growth in capital or labour

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– forecast
– explain

– to estimate potential GDP, need:
  – trend estimate for labour & capital
    $(\Delta L/L)$ $(\Delta K/K)$
  – A, treated as exogenous

– alternative method/labour productivity growth rate accounting method

growth rate in potential GDP = long-term growth rate of labour force + long-term growth rate in labour productivity

Further extensions/

$$Y = AF(N, L, H, K_{IT}, K_{NT}, K_P)$$

N – natural resources
L – Labour
H – quality of Labour (human capital)
K_{IT} – ICT Capital
K_{NT} – non-ICT Capital
K_P – public capital

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- forecast
- explain

⇒ Natural Resources/
1) renewable resources
2) non-renewable resources
- access to resources is important (through trade)
- ownership & production of natural resources is not necessary for a country to achieve a high level of income
- may even restrain growth
  - may simply fail to develop the economic institutions to support growth
  - strong export demand for their resources may drive up the currency and make other sectors uncompetitive

⇒ Labour Supply/
- labour force = working age population that is either employed or available for work
  - Depends on
    - population growth
    - labour force participation & avg. hrs. worked
    - net migration

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- forecast
- explain

⇒ Labour Supply/
1) population growth – specifically working age population (16-65) – age mix important
   - lower in developed countries
2) Labour force participation – %age of working age population in the labour force
   - has been increasing (more women entering)
- pop. growth can ↑ output but not necessarily ↑ output/capita
- labour force participation does both
3) Net migration Note: boosts potential GDP
   - if a country is running a persistent output gap $[(y - y^*) < 0]$, increased immigration will make it worse and increase unemployment

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- explain

⇒ Labour Supply/
4) Average hours worked - has been trending lower
⇒ Labour Quality/ - knowledge & skills from education, training, experience
- better educated/more skilled workers more productive & more adaptable (also earn higher wages)
- increased through investment in education
- tends to also increase the pace of technological change
⇒ ICT & non-ICT Capital/
• physical stock increases each year as long as net investment is positive
- correlation between growth & investment is high
- ICT Capital - offer network externalities
- capital deepening + technological progress (TFP)

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- explain

⇒ ICT & non-ICT Capital/
• non-ICT ⇒ Capital Deepening
⇒ Technology/ most important factor affecting growth of per capita GDP
- technology results in an upward shift of the production function
- produce more and/or higher quality goods & services with the same resources or inputs
⇒ not mentioned: allows same output with fewer resources or inputs
Issue: Developed economy
- technological advancement increases potential GDP but may also potentially increase unemployment
- net immigration increase potential GDP but may also make unemployment worse

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- forecast
- explain

⇒ Technology/
- public & private R&D
- embodied in capital goods (new machinery, equipment, software)
- the state of technology in captured in A (in the prod. function)
- level of labour productivity – higher in developed countries
- growth rate of labour productivity – higher in developing countries

⇒ Public Infrastructure/
- boosts productivity of private investment

Growth Theories

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LOS i
- compare

⇒ Classical Model/ (Thomas Malthus)
- growing population, limited resources

⇒ leads to overpopulation, depletion of resources
(Collapse, Jared Diamond)

$Y = F(L, land)$

- when income/capita > subsistence level

(exogenous)
(temporary)

technological progress + land expansion = higher population growth
(labour productivity)

diminishing marginal returns + higher population

income/capita falls back to subsistence level

- constant standard of living over time
- new technology = larger population, not richer population

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- compare

⇒ **Classical Model/ (Thomas Malthus)**

failures 1) as income/capita increases, population growth slows, not accelerates

2) growth rate of technological process > rate of diminishing returns

⇒ **Neo-Classical Growth Model/ (Solow growth model)**

$Y = AF(K,L) = AK^{\alpha}L^{1-\alpha}$

Y/capita – rate of tech. change – pop. growth – savings/investment rate

equilibrium = steady state rate of growth

$\underbrace{\text{steady state}}_{Y/K = \text{constant}}$

$\therefore$ $\underset{k}{K/L}$ & $\underset{y}{Y/L}$ grow at the same rate

Page 24
LOS i
- compare

⇒ **Neo-Classical Growth Model/** $y = Y/L$ output per worker $k = K/L$ capital per worker

⇒ rate of change of k $\Delta k/k = \Delta K/K - \Delta L/L$

⇒ rate of change of y $\Delta y/y = \Delta Y/Y - \Delta L/L$

or $\Delta y/y = \Delta A/A + \alpha\,\Delta k/k$ $(y = Ak^{\alpha})$

- What is ΔK?
  - K increases by I & decreases by depreciation

$\underset{\text{saving rate}}{I = sY}$ - $\underset{\text{depreciation rate}}{\delta K}$

$\therefore\ \Delta K = sY - \delta K$

divide by K $\dfrac{\Delta K}{K} = \dfrac{sY}{K} - \delta$ $\therefore\ \Delta k/k = \dfrac{sY}{K} - \delta - \underset{(n)}{\Delta L/L}$

Page 25
LOS i
- compare

⇒ **Neo-Classical Growth Model/** - in steady state,
growth rate of capital/worker = growth rate of output/worker

$$\Delta k/k = \Delta y/y = \Delta A/A + \alpha\, \Delta k/k$$

$$\Delta k/k - \alpha\, \Delta k/k = \Delta A/A$$

$$\Delta k/k\,(1-\alpha) = \Delta A/A$$

$$\Delta y/y = \Delta k/k = \frac{\Delta A/A}{(1-\alpha)}$$

let $\theta = \Delta A/A$ → growth rate of TFP

$$\Delta y/y = \frac{\theta}{1-\alpha}$$ growth rate of output/capita — steady state growth rate of labour productivity

$$\Delta Y/Y = \frac{\theta}{1-\alpha} + \Delta L/L$$ growth rate of output

Page 26
LOS i
- compare

⇒ **Neo-Classical Growth Model/**

- last step ⇒ substitute $\left(\frac{\theta}{1-\alpha}\right)$ into $\Delta k/k = sY/K - \delta - \Delta L/L$

$$\frac{\theta}{(1-\alpha)} = \frac{sY}{K} - \delta - \Delta L/L$$

$$\frac{\theta}{(1-\alpha)} + \delta + \Delta L/L = sY/K$$

$$\left(\frac{1}{S}\right)\left[\left(\frac{\theta}{1-\alpha}\right) + \delta + \Delta L/L\right] = Y/K \quad \text{or} \quad Y/K = \left(\frac{1}{S}\right)\left[\left(\frac{\theta}{1-\alpha}\right) + \delta + \Delta L/L\right] = \Psi$$

So/ in steady state

Y/K is constant, k & y both grow at $\left(\frac{\theta}{1-\alpha}\right)$

- on steady state growth path

marginal product of capital = $\alpha(Y/K) = r$ (real interest rate)

result: in steady state, capital deepening has no effect on:

① growth rate of economy ② MPK

Page 27
LOS i
– compare

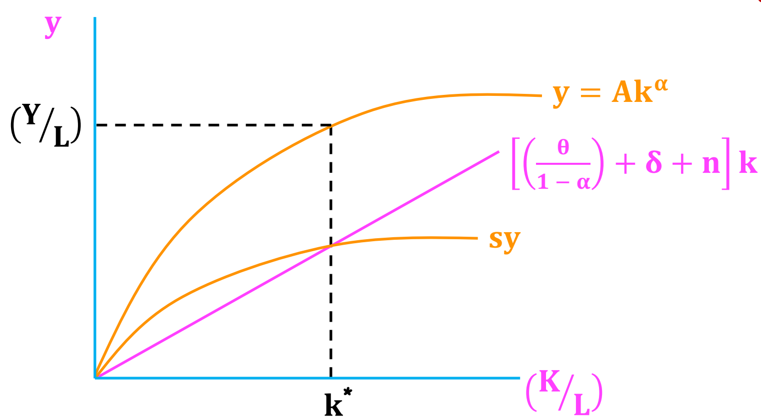
⇒ Neo-Classical Growth Model/

– steady state equilibrium occurs when:

$$S_y = \left[\underbrace{\left(\frac{\theta}{1-\alpha}\right) + \delta + n}_{}\right] k$$

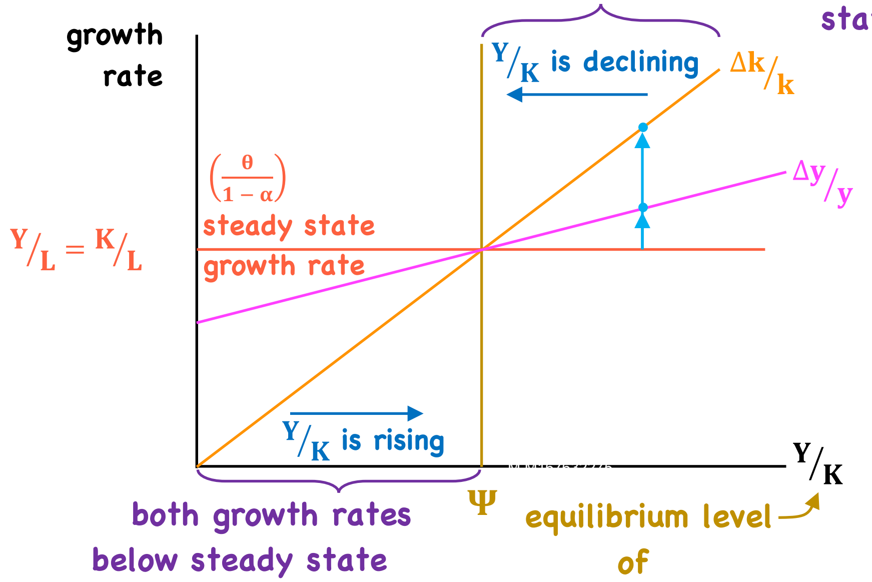
saving/worker = investment required to keep K constant

investment:
- capital for new workers entering the workforce (n)
- replace PPE (δ)
- keep MPK = r $\left(\frac{\theta}{1-\alpha}\right)$



Page 27b
LOS i
– compare

⇒ Neo-Classical Growth Model/ both growth rates > steady state



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LOS i
- compare

⇒ Neo-Classical Model/

1. Capital Accumulation
   - affects level of output, not growth rate
   - economies move towards a steady state
   - in the steady state, growth rate of output does not depend on capital accumulation

2. Deepening vs. Technology
   - growth slows as capital is added
   - capital deepening limits out
   - without increases in TFP, growth of y will slow

3. Convergence
   - growth rate of developing countries should exceed that of developed countries
   - per capita incomes should converge

4. Effect of s on Growth
   - s increases growth rate until a steady state is reached
   - in steady state, growth does not depend on s
   - countries w/ higher s, have higher y, k, & TFP

Page 29a
LOS i
- compare

⇒ Endogenous Growth Theory/ (Paul Romer/86)

- technological progress is not exogenous
  (models attempt to explain tech. progress)

- economy does not necessarily converge to a steady state of growth
  - no diminishing marginal returns to capital for the economy as a whole (allows for increasing returns to scale)

⇒ increasing s permanently increase rate of economic growth

- definition of capital is broadened to include $\underbrace{\text{knowledge/R\&D}}_{\text{ideas}}$

(partially excludable)

- large externalities

Page 29
LOS i
- compare

⇒ Endogenous Growth Theory/

$y_e = f(k_e) = ck_e$ e - endogenous

output/worker is proportional to capital/worker
(c = constant MPK)

• output/worker grows at the same rate as capital/worker

⇒ a higher saving rate implies a permanently higher growth rate

LOS k
- describe

- if private sector under-invests in R&D, direct government spending and/or R&D tax breaks/subsidies may correct the situation

- higher level of spending on knowledge capital could translate into faster economic growth <u>even in the long-run</u>

= further ⇒ no reason why incomes of developed & developing countries should converge

Convergence

Page 30
LOS j
- explain
- evaluate

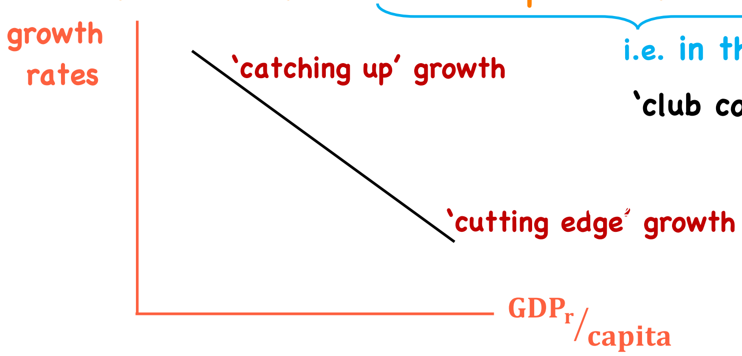
- countries with low per capita incomes should grow faster than those with high levels

2 types of convergence

1) absolute convergence - developing countries will eventually catch up with developed countries w.r.t. per capita output

2) conditional convergence - convergence is conditional on institutions & $\underbrace{\text{certain input factors being similar}}_{\text{i.e. in the same club}}$

'club convergence'



Page 31
LOS j
- explain
- evaluate

- 2 routes to convergence
  1) capital accumulation & capital deepening
  2) imitation/adoption of technology

Endogenous/ makes no prediction that convergence should occur

- empirical evidence/
  - poorer countries are diverging
    - could converge if they adopt the appropriate legal, political & economic institutions
  - club countries are converging

Growth in an Open Economy

Page 32
LOS l
- describe

- investment not constrained by domestic savings
  - s increases, capital investment increases
- resources shift to comparative advantage sectors
  - efficiency ↑, profits ↑, overall productivity ↑
- access to larger markets
- can import technology, increasing rate of technological progress
  - per capita incomes rise
  - growth rate of output will rise

REVIEW

Currency Exchange Rates

Review - 1

⇒ Quoting Conventions/ B.P or $^{P}/_{B}$ e.g./ USD.CAD or $^{CAD}/_{USD}$

1) Dealer bid < Interbank offer

Dealer offer > Interbank bid

Interbank Market
bid — Dealer — offer
bid — Client — offer

→ bid–offer spread depends on
- pair involved (major / cross)
- time of day
  - Japan • Europe • N.A.
- volatility of pair

2 arbitrage constraints

2) Triangular arbitrage/ dealer crossrate bid < Implied Interbank crossrate offer

$$\frac{USD}{EUR} \times \frac{EUR}{JPY} = \frac{USD}{JPY}$$

1.0522/24 118.10/15 124.26/34

$$\frac{A}{B} \times \frac{C}{A} = \frac{C}{B}$$

$$\frac{A}{B} + \frac{A}{C} = ?$$

↳ invert to $^{C}/_{A}$

new bid = $\frac{1}{\text{offer}}$

new offer = $^{1}/_{\text{bid}}$

Review - 2

⇒ Forward Rates/ i_d – domestic r_f } Libor rates
i_f – foreign r_f

$$^{P}/_{B} = {}^{f}/_{d}$$

$$(1+i_d) = S_{f/d}(1+i_f)\left(\frac{1}{F_{f/d}}\right) \Rightarrow F_{f/d} = S_{f/d}\underbrace{\frac{(1+i_f)}{(1+i_d)}}$$

$i_f > i_d$ – forward premium

$$\underbrace{F_{f/d} = S_{f/d}\left(\frac{1+i_f[^{act}/_{360}]}{1+i_d[^{act}/_{360}]}\right)}$$

covered interest rate parity

(must hold)

$$F_{P/B} = S_{P/B}\left(\frac{1+i_P[^{act}/_{360}]}{1+i_B[^{act}/_{360}]}\right)$$

⇒ Points/ forward rates are quoted in terms of points

$S_{P/B}$	1.3642/1.3644
1-mos.	3.1/3.3
3-mos.	15.6/16.2
6-mos.	37/38.7
12-mos.	96/98.9

$$\underbrace{F_{f/d} - S_{f/d}}_{\text{for. pr./disc.}} = S_{f/d}\left[\frac{^{act}/_{360}}{1+i_d(^{act}/_{360})}\right](i_f - i_d)$$

(pg. 9 – bottom)

Spot vs. Forward Rates

Page 9
LOS c
- distinguish
- calculate

⇒ domestic currency will trade at a forward:

premium $\left(F_{f/d} > S_{f/d}\right)$ if $i_f > i_d$ $(i_f - i_d)$

discount $\left(F_{f/d} < S_{f/d}\right)$ if $i_f < i_d$

- now lets put $^f/_d$ back into $^P/_B$: $F_{P/B} = S_{P/B}\left(\frac{1 + i_P[Act/360]}{1 + i_B[Act/360]}\right)$

e.g./ $S_{CAD/AUD} = 1.0145$

270 day Libor AUD = 4.87%

270 day Libor CAD = 1.41%

forward premium/discount?

$$F_{P/B} - S_{P/B} = S_{P/B}\left(\frac{Act/360}{1 + i_B[Act/360]}\right)(i_P - i_B)$$

$$= 1.0145\left(\frac{270/360}{1 + .0487(270/360)}\right)(.0141 - .0487)$$

$$= -.0254$$

Currency Exchange Rates

Review - 3

⇒ Mark-to-Market/

AUD/GBP 1.6210/1.6215

3-mos. pts.
130/40

bought £10M for $^{AUD}/_{GBP}$ = 1.61

want to close position

(1.6340 - 1.6100) × £10M
= 240k AUD

$$\frac{240,000}{(1 + i_{AUD}[act/360])}$$

(pg. - 14 e.g.)

Note: if contract is being rolled forward - called an FX-Swap
- use mid-point of bid-offer
(common spot rate applied to both legs)

⇒ Parity Relations/

1) Covered Interest Rate Parity - must hold

- supported where capital is permitted to flow freely, spot/forward markets are liquid, financial markets are stress-free

Review - 4

⇒ Parity Relations/

2) Uncovered Interest Rate Parity - UIRP

$\underbrace{i_f - i_d}_{\text{interest rate differential}} = \%\Delta S^e_{f/d}$

- expected return on an unhedged forex investment should equal the return on a comparable domestic currency investment

- country w/ higher i is expected to see its currency depreciate

(implies no return to carry trades)

- does not hold (short & medium term)

3) PPP - Purchasing Power Parity/

- relationship between fx. rates and inflation differentials

$P^x_f = S_{f/d} P^x_d$ - price of x

e.g./ P^x_{EUR} = €100 $S_{USD/EUR}$ = 1.4000

⇒ then P^x_{USD} = $140

$S_{f/d} = P_f / P_d$ - general price level

(absolute version of PPP)

Review - 5

⇒ Parity Relations/

3) PPP - Purchasing Power Parity/ - assumes all domestic & foreign goods are tradable and that price indices are identical

relative/ $\%\Delta S_{f/d} \approx \underbrace{\pi_f - \pi_d}_{\text{inflation differential}}$ - currency of country with high inflation should depreciate

ex-ante/ $\%\Delta S^e_{f/d} \approx \pi^e_f - \pi^e_d$ - expected changes in spot fx. rates driven by expected differences in inflation rates

4) Fisher Effect & Real Interest Rate Parity/

- combines both - interest rate differentials
- inflation differentials

$$\left.\begin{array}{l} i_d = r_d + \pi^e_d \\ i_f = r_f + \pi^e_f \end{array}\right\}$$ nominal rate = real rate + inflation
(i) (r) (π)

Review - 6

⇒ Parity Relations/

4) Fisher Effect & Real Interest Rate Parity/

$$i_f - i_d = \underbrace{(r_f - r_d)}_{\text{yield spread}} + \underbrace{(\pi_f^e - \pi_d^e)}_{\text{inflation differential}}$$

$$(r_f - r_d) = (i_f - i_d) - (\pi_f^e - \pi_d^e)$$

$$(r_f - r_d) = \%\Delta S^e_{f/d} - \%\Delta S^e_{f/d} = 0$$

(from UIRP) (from PPP)

⇒ real interest rate parity

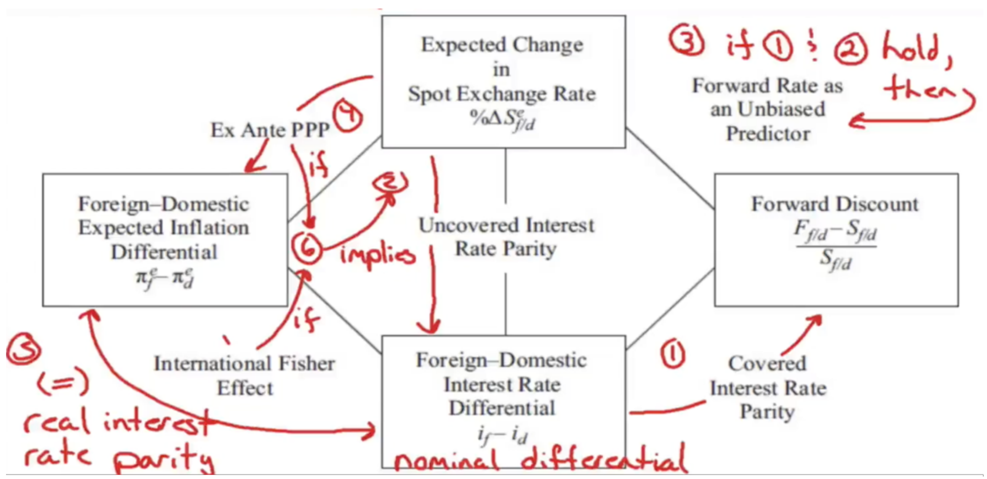
International Fisher Effect: $i_f - i_d = \pi_f^e - \pi_d^e$

(pg.-25)

Page 25
LOS e, f, g

- if all International parity conditions held:

$\%\Delta S^e_{f/d}$ = forward premium/discount = $i_f - i_d = \pi_f^e - \pi_d^e$
(%'age)



- if all conditions held, it would be impossible for a global investor to earn consistent profits on currency movements

Review - 7

⇒ Carry Trade/ lend in high yielding currencies

borrow in $\underbrace{\text{low yielding currencies}}_{\text{funding currencies}}$ (pg.-31 e.g.)

⇒ Balance of Payments Impact/

Capital Acct. → financial flows - usually dominant factor in fx. rate movements

Current Acct. → real g/s flows

- prices of g/s adjust slower
- production of g/s takes time

A) Current acct. imbalances

1) flow supply/demand channel

trade surplus - more demand for currency - upward pressure

trade deficit - more selling of currency - downward pressure

- adjustment required depends on:

a) gap between X/M b) response of X/M prices to Δfx. rate

c) " " " demand to Δfx. rate

Carry Trade

Page 31
LOS i
- describe
- calculate

	1-yr. Libor	Pair	1-yr. later
JPY	0.10%	JPY/USD → 81.30	80.00
AUD	4.50%	USD/AUD → 1.0750	1.0803

- borrow in yen, invest in 1-yr. AUD Libor
- after 1 yr., calculate the all-in return to this trade

$$^{JPY}/_{AUD} = {}^{JPY}/_{USD} \times {}^{USD}/_{AUD}$$

original

81.30 × 1.0750 = 87.40

1-yr. later

80.00 × 1.0803 = 86.42

$$\text{All-in} = \frac{86.42}{87.40}(1.0450) - .001$$

$$= 1.033283 - .001$$

$$= 1.032283 \sim 3.23\%$$

Currency Exchange Rates

Review - 7

⇒ Carry Trade/ lend in high yielding currencies

borrow in $\underbrace{\text{low yielding currencies}}_{\text{funding currencies}}$ (pg.-31 e.g.)

⇒ Balance of Payments Impact/

Capital Acct. → financial flows - usually dominant factor in fx. rate movements

Current Acct. → real g/s flows

- prices of g/s adjust slower
- production of g/s takes time

A) Current acct. imbalances

1) flow supply/demand channel

trade surplus - more demand for currency - upward pressure

trade deficit - more selling of currency - downward pressure

- adjustment required depends on:

a) gap between X/M

b) response of X/M prices to Δfx. rate

c) " " " demand to Δfx. rate

Review - 8

⇒ Balance of Payments Impact/

A) Current acct. imbalances

2) portfolio balance channel - rebalancing fx. reserves

3) debt sustainability channel - persistent trade deficits result in increasing debt → major dep. may be necessary

B) Capital acct. (capital flows)

1) Real Interest Rate Diff., Cap. flows & Fx. Rate

capital responds to int. rate diff., risk premia, & fx. rate expectations

$$q_{f/d} = \bar{q}_{f/d} + [\underbrace{(r_d - r_f)}_{\text{real int. rate diff.}} - \underbrace{(\varphi_d - \varphi_f)}_{\text{relative risk premia}}]$$

2) Int. Rate diff., Carry Trades & Fx. Rate

will rise when

$$q_{L/H} = \bar{q}_{L/H} + \underbrace{(i_H - i_L)}_{\text{yield spread widens}} - \underbrace{(\pi_H^e - \pi_L^e)}_{\text{inflation exp. spread}\downarrow} - \underbrace{(\varphi_H - \varphi_L)}_{\text{risk premia spread}\downarrow}$$

Review - 9

⇒ Balance of Payments Impact/

B) Capital acct. (capital flows)

3) Equity Market Trends & FX rates - not a stable relationship

⇒ Monetary & Fiscal Policy/

a) Mundell-Fleming Model/

Expansionary

Monetary Policy	Fiscal Policy
- reduce interest rates	- increase spending/lower taxes
- leads to capital outflows	- exerts upward pressure on interest rates
- downward pressure on fx. rates	- leads to capital Inflows
	- upward pressure on fx. rates

- more responsive capital flows are
  - larger the fx. dep.
  - larger the fx. appr.

Review - 10

⇒ Monetary & Fiscal Policy/

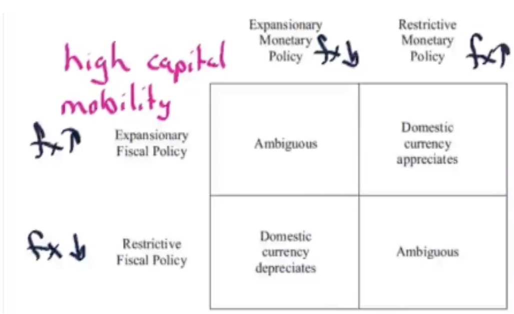
a) Mundell-Fleming Model/

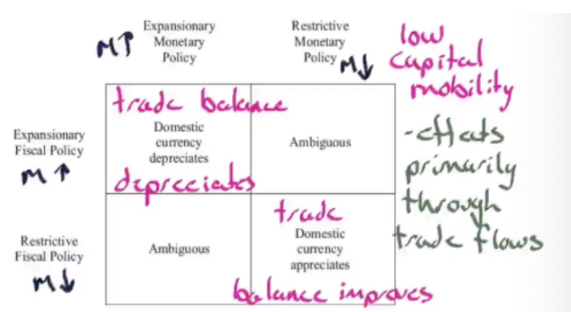
w/ fixed rates

central bank will have to buy its own currency (reduce money held by others)	central bank will have to sell its own currency
- offsets exp. policy (limited by stock of reserves)	- supports exp. policy (theoretically unlimited)

a) pursue independent monetary policy
b) permit capital to flow freely
c) defend fx. rate

cannot do all 3
- typically use capital controls





Review - 11

⇒ Monetary & Fiscal Policy/
b) Monetary Approach

w/ flexible prices monetary policy → price level/ rate of inflation → fx. rates
- assumes PPP holds (constant real fx. rate)
$[\uparrow M, \uparrow CPI, \downarrow S_{f/d}]$

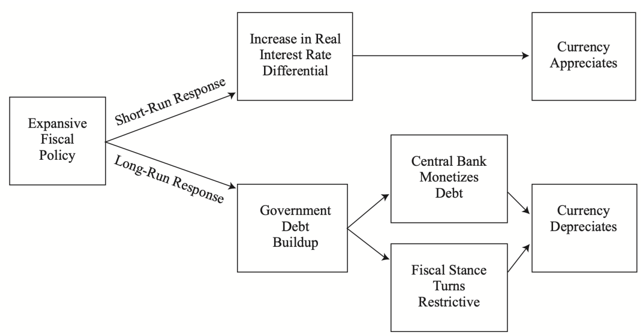
w/ somewhat flexible prices Dornbusch Overshooting Model
- prices have limited flexibility in S.R.
- fully flexible in L.R. $\underbrace{(\uparrow M, \uparrow CPI, \downarrow \text{fx. rate})}_{\text{long-run}}$

$\underbrace{\uparrow M, \text{CPI-unch.}}_{\text{short-run}}$

Review - 12

⇒ Monetary & Fiscal Policy/
d) Fiscal policy

combine MF + portfolio balance approach ⇒



⇒ Intervention/
a) undervalued currency
- no intervention
b) fairly - overvalued currency, no inflation threat
- unsterilized intervention - expand M (slow down money flows)
c) b) + inflation - sterilized intervention
- sell domestic securities to private sector
d) limit to sterilization - tighten fiscal policy

- if intervention fails ⇒ capital controls

Economic Growth

Review – 1

• actual vs. $\underbrace{\text{potential}}_{\text{LR aggregate output}}$

- growth rate = economy's sustainable g
- identify factors that support LR economic growth (i.e. potential GDP)

⇒ Comparing Performance/ GDP_r/capita

- to compare, convert each country's real GDP to a common currency at a $\underbrace{\text{PPP exchange rate}}_{\text{should measure material well-being}}$

(Note: do not use market fx. rates)

- developed economies → high GDP/capita
- developing economies → low GDP/capita

⇒ Performance Enhancing Factors/

1) Savings & Investment
2) Financial Markets & Intermediaries

Review – 2

⇒ Performance Enhancing Factors/

3) Political Stability, Rule of Law, Property Rights
4) $\underbrace{\text{Education}}_{\text{focus of innovation}}$ & Health Care Systems

focus of innovation:

leading edge
- investments in post-secondary

imitation
- investments in primary & secondary

5) Tax & Regulatory Systems
6) Free Trade & Unrestricted Capital Flows

FDI & indirect inv.

⇒ Stock Market vs. Economy/ - in the long-run, earnings growth rate cannot exceed the growth rate of potential GDP

$$\underbrace{\%\Delta \text{ in stock market value} = \%\Delta GDP}_{\text{long run}} + \underbrace{\%\Delta \text{ in share of earnings in GDP} + \%\Delta \text{ in the P/E multiple}}_{= 0 \quad \text{long-term} \quad = 0}$$

- the same factors that limit economic growth limit aggregate economic growth

Review - 3

⇒ Fixed Income/ - pot. GDP helps gauge inflationary pressure

- higher g in y^* improves the general credit quality of issuers

$\underbrace{(y - y^*)\text{ - output gap}}$ determinant of real interest rates

- as $y^*\uparrow$, inflationary pressure drops

⇒ Simple Production Function/

$$Y = AF(K, L) \qquad A \rightarrow \text{TFP}$$

Cobb-Douglas $\quad F(K, L) = \underbrace{K^{\alpha}}L^{1-\alpha}$ $\quad \alpha$ - capital's share of GDP

paid their marginal product

- profit max: MPK - rental price of capital (r) - real int. rate

MPL - real wage rate

$$MPK = \frac{\alpha Y}{K} = r \qquad \alpha = \frac{rK}{Y}$$ - share of GDP paid to capital

$$MPL = \frac{(1-\alpha)Y}{L} \qquad (1-\alpha) = \frac{Lw_r}{Y}$$ - share of GDP paid to Labour

Review - 4

⇒ Simple Production Function/

Cobb-Douglas (2 properties)

1) constant returns to scale

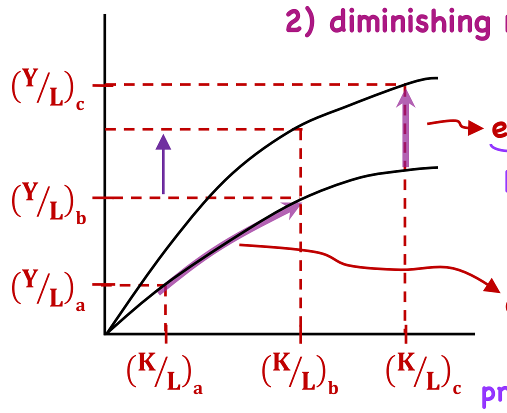
$$Y/L = A(K/L)^{\alpha}(L/L)^{1-\alpha}$$

output per worker - y

capital-to-labour ratio

$= Ak^{\alpha}$ → the amount of goods a worker can produce depends on the amount of capital for each worker, TFP, & the share of capital in GDP

2) diminishing marginal productivity



→ effect of increase in TFP - a source of g even in steady state

prime source of g in developed economies (tech. progress)

effect of capital deepening

prime source of g in developing economies (MPK > r)

→ steady state (MPK = r)

Cap. Deep. no longer a source of g.

Review - 5

⇒ Growth Accounting/

$$\frac{\Delta Y}{Y} = \frac{\Delta A}{A} + \alpha\frac{\Delta K}{K} + (1-\alpha)\frac{\Delta L}{L}$$

g rate of output; rate of tech. ch. (exogenous); g. rate of capital; - g rate of Labour

- Y, K, L available from GDP accts.
- A must be estimated
- $\alpha \sim .3$ for US

- to forecast potential GDP, use trend estimate of labour $(\Delta L/L)$ & capital $(\Delta K/K)$

or/ Labour productivity growth rate accounting method/

g rate in pot. GDP = Long-Term g rate in labour force + L.T g rate in labour productivity

= Further extensions/

$$Y = AF(N, L, H, K_{IT}, K_{NT}, K_P)$$

N: nat. res. (access vs. ownership)
L: quant. of L
H: qual. of L
K_{IT}: ICT
K_{NT}: non-ICT
K_P: public

L depends on
- pop. g.
- Labour force participation
- net migration

Review - 6

⇒ Classical Model/ Malthus   Y = F(L,Land)

Y/L > subsistence level → population growth

∴ constant standard of living over time
(new tech. = larger population, not richer population)

⇒ Neo-Classical Model/ Solow growth model

$$Y = AF(K, L) = AK^{\alpha}L^{1-\alpha}$$

↓

equilibrium = steady state rate of g.   i.e. Y/K constant
k & y grow at same rate

⇒ Endogenous Growth Theory/ Romer
- tech. progress is not exogenous
- economy does not necessarily converge to a steady state of growth
- no diminishing returns to capital (even increasing returns to scale)
- definition of capital now includes knowledge

Growth Theories

⇒ Neo-Classical Growth Model/ $y = Y/L$ output per worker $k = K/L$ capital per worker

Page 24
LOS i
- compare

⇒ rate of change of k $\Delta k/k = \Delta K/K - \Delta L/L$

⇒ rate of change of y $\Delta y/y = \Delta Y/Y - \Delta L/L$

or $\Delta y/y = \Delta A/A + \alpha\, \Delta k/k$ $(y = Ak^{\alpha})$

- What is ΔK?
  - K increases by I & decreases by depreciation

$I = sY$ - δK

saving rate   depreciation rate

$\therefore\ \Delta K = sY - \delta K$

divide by K $\dfrac{\Delta K}{K} = \dfrac{sY}{K} - \delta$ $\therefore\ \Delta k/k = \dfrac{sY}{K} - \delta - \Delta L/L$ (n)

⇒ Neo-Classical Growth Model/ - in steady state, growth rate of capital/worker = growth rate of output/worker

Page 25
LOS i
- compare

$\Delta k/k = \Delta y/y = \Delta A/A + \alpha\, \Delta k/k$

$\Delta y/y = \Delta k/k - \alpha\, \Delta k/k = \Delta A/A$

$\Delta y/y = \Delta k/k\,(1-\alpha) = \Delta A/A$

$\Delta y/y = \Delta k/k = \dfrac{\Delta A/A}{(1-\alpha)}$ let $\theta = \Delta A/A$ → growth rate of TFP

$\Delta y/y = \dfrac{\theta}{1-\alpha}$ growth rate of output/capita   steady state growth rate of labour productivity

$\Delta Y/Y = \dfrac{\theta}{1-\alpha} + \Delta L/L$ growth rate of output

Page 26
LOS i
- compare

⇒ Neo-Classical Growth Model/

- last step ⇒ substitute $\left(\frac{\theta}{1-\alpha}\right)$ into $\Delta k/_k = sY/_K - \delta - \Delta L/_L$

$$\frac{\theta}{(1-\alpha)} = \frac{sY}{K} - \delta - \Delta L/_L$$

$$\frac{\theta}{(1-\alpha)} + \delta + \Delta L/_L = sY/_K$$

$$\left(\frac{1}{S}\right)\left[\left(\frac{\theta}{1-\alpha}\right) + \delta + \Delta L/_L\right] = Y/_K \quad \text{or} \quad Y/_K = \left(\frac{1}{S}\right)\left[\left(\frac{\theta}{1-\alpha}\right) + \delta + \Delta L/_L\right] = \Psi$$

So/ in steady state

$Y/_K$ is constant, k & y both grow at $\left(\frac{\theta}{1-\alpha}\right)$

- on steady state growth path

marginal product of capital = $\alpha(Y/_K) = r$ (real interest rate)

result: in steady state, capital deepening has no effect on:

① growth rate of economy ② MPK

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LOS i
- compare

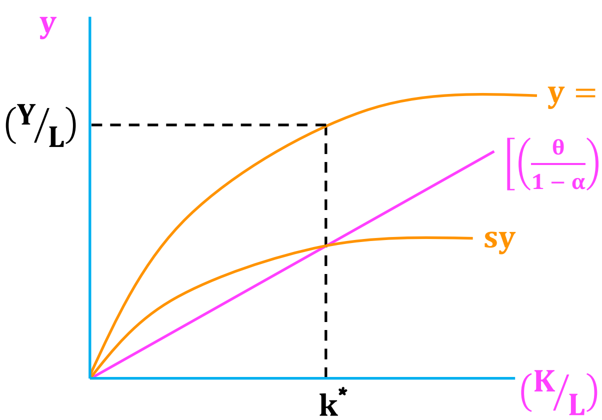
⇒ Neo-Classical Growth Model/

- steady state equilibrium occurs when:

$$\underset{\downarrow \text{ saving/worker}}{S_y} = \left[\underbrace{\left(\frac{\theta}{1-\alpha}\right) + \delta + n}_{\text{investment required to keep } K \text{ constant}}\right]k$$

investment:

- capital for new workers entering the workforce (n)
- replace PPE (δ)
- keep MPK = r $\left(\frac{\theta}{1-\alpha}\right)$



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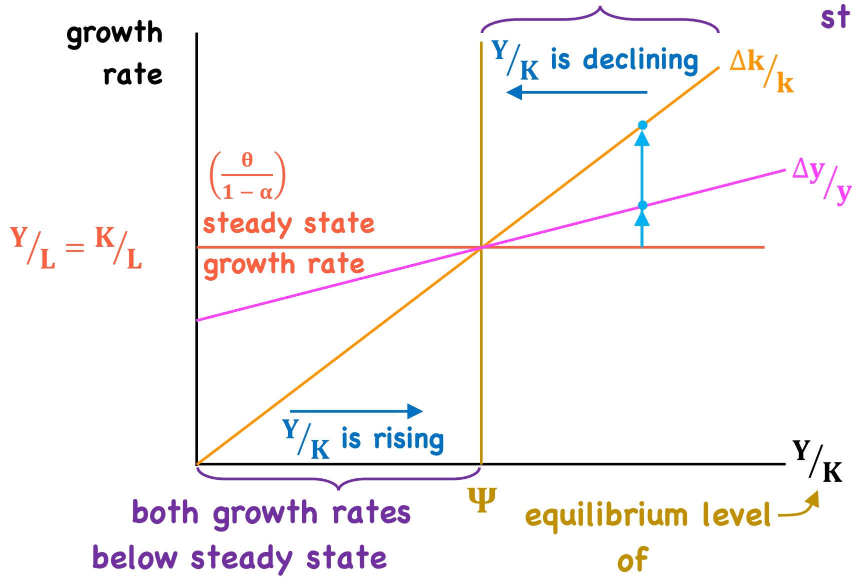
⇒ Neo-Classical Growth Model/ both growth rates > steady state



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⇒ Neo-Classical Model/

1. Capital Accumulation
   - affects level of output, not growth rate
   - economies move towards a steady state
   - in the steady state, growth rate of output does not depend on capital accumulation
2. Deepening vs. Technology
   - growth slows as capital is added
   - capital deepening limits out
   - without increases in TFP, growth of y will slow
3. Convergence
   - growth rate of developing countries should exceed that of developed countries
   - per capita incomes should converge
4. Effect of s on Growth
   - s increases growth rate until a steady state is reached
   - in steady state, growth does not depend on s
   - countries w/ higher s, have higher y, k, & TFP

Economic Growth

Review - 6

⇒ Classical Model/ Malthus $Y = F(L, Land)$

Y/L > subsistence level → population growth

∴ constant standard of living over time

(new tech. = larger population, not richer population)

⇒ Neo-Classical Model/ Solow growth model

$Y = AF(K, L) = AK^{\alpha}L^{1-\alpha}$

↓

equilibrium = steady state rate of g. i.e. Y/K constant

k & y grow at same rate

⇒ Endogenous Growth Theory/ Romer

- tech. progress is not exogenous
- economy does not necessarily converge to a steady state of growth
- no diminishing returns to capital

(even increasing returns to scale)

- definition of capital now includes knowledge

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⇒ Endogenous Growth Theory/ Romer

- a higher saving rate implies a permanently higher g rate
- no reason why incomes of developed & developing economies should converge

⇒ Convergence/ - countries w/ low incomes/capita should grow faster

a) absolute convergence - developing countries will eventually catch up with developed w.r.t. per capita output

b) conditional convergence - conditional on institutions & certain input factors being similar

- 2 routes to convergence

1) capital deepening/capital accumulation

2) imitation/adoption of technology

Review - 8

⇒ Growth in an open economy/

1) investment not constrained by domestic saving
2) resources shift to comparative advantage sectors
3) access to larger markets
4) can import technology, increasing rate of technological progress